Patryk Kozlowski
PhD Candidate, Chemistry & Chemical Biology
Harvard University
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December 5, 2025

Dear Members of the Dudley Community Resident Tutor Selection Committee,

I am a PhD candidate in Chemistry and Chemical Biology at Harvard, and I am writing to express my interest in joining the Dudley Community community as a Resident Tutor. My professional background is as a researcher and mentor. My personal background is as a stroke and cancer survivor. These experiences have imbued in me a unique perspective that will allow me to support the students at Dudley Community in their academic and personal growth.

I have developed an expertise in the practice and communication of science. I have spent most of my life as a researcher in quantum chemistry; this field is highly interdisciplinary, with a foot in physics, chemistry, computer science, and applied mathematics. Additionally, I am able to operate outside my area of expertise. As a member of the Harvard Energy Journal Club, I have given presentations on the energy demands of AI, carbon taxation, and climate tipping points; as a Teaching Fellow for Physical Sciences 11, I have learned to relay complex chemistry concepts to students from diverse academic backgrounds. These experiences have prepared me to mentor students pursuing all kinds of scientific disciplines at Dudley Community.

I am uniquely qualified to support students in their personal journeys. After I suffered a stroke four years ago, I could not move in my hospital bed, but now I am pursuing a PhD at Harvard. This would not have been possible without learning how to manage time effectively. Every week, I interpolate my work hours between programming and doing physical therapy exercises. This experience has prepared me to support students in achieving their personal definitions of success.

I am drawn to Dudley Co-op due to its independent spirit. I would love to work with students at the Co-op who have chosen a nonconforming lifestyle.

Thank you for considering my application and I would welcome the opportunity to discuss how my experience and values align with the House's mission.

Sincerely,
Patryk Kozlowski

PATRYK KOZLOWSKI

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EDUCATION

Harvard University, Cambridge, MA — *PhD Candidate, Chemistry & Chemical Biology*
Expected: June 2029

California Institute of Technology (Caltech), Pasadena, CA — *BS in Chemistry*
GPA: 3.74/4.0 • *June 2024*

QUANTUM CHEMISTRY EXPERIENCE

Research Fellow — *Harvard University* (Advisor: Prof. Joonho Lee)

July 2024 – Present

- Developing Green's function methods for periodic quantum chemistry.
- Exploring **open quantum system** ideas to enable accurate and efficient simulations of correlated materials.

Research Fellow (Senior Thesis) — *Caltech* (Advisor: Prof. Garnet Chan)

Oct 2023 – June 2024

- Implemented GW, a Green's function method, for molecules.
- **Publications & Presentations:**
  - Deposited thesis
  - Presented at Caltech Senior Thesis & Goldwater Symposiums

Research Fellow — *Caltech* (Advisor: Prof. Garnet Chan)

Nov 2022 – Aug 2023

- Implemented and optimized the **Full Configuration Interaction (FCI)** method for a chain of 6 hydrogen atoms.

John Stauffer SURF Fellow — *Caltech* (Advisor: Prof. Garnet Chan)

June 2020 – Sept 2020

- Computed platinum surface energies using coupled-cluster theory.
- **Publications & Presentations:**
  - Oral presentation, Caltech SURF Seminar Day
  - Article in *Caltech Undergraduate Research Journal*

John Stauffer SURF Fellow — *Caltech* (Advisor: Prof. Ryan Hadt)

June 2019 – Sept 2019

- Developed computational model for **spin-phonon coupling** in Co(III) complexes.
- **Publications:**
  - Presentation at Caltech SURF Seminar Day
  - Co-author on *Journal of Chemical Physics* publication

BROAD SCIENCE EXPERIENCE

Member — *Harvard Energy Journal Club*

Sept 2024 – Present

- Participate in a weekly decarbonization-focused journal club discussions, primarily composed of GSAS affiliates.
- Presented on **"Decarbonizing AI" (Nov 2024)**, **The Carbon Tax (Apr 2025)**, and **Tipping Points (Nov 2025)**.

Teaching Fellow — Physical Sciences 11 — *Harvard University*

Jan 2025 – May 2025

- Supported instruction in *Physical Sciences 11*, an introductory physical sciences course for Harvard undergraduates.
- Led weekly discussion sections and office hours.

Associate Editor — *Caltech Undergraduate Research Journal*

Dec 2023 – May 2024

- Prepared an article on the nonlinear spectroscopy of cuprates for publication.

Teaching Assistant — Intro Quantum Mechanics (Chemistry) — *Caltech*

Oct 2020 – Dec 2020

- Held virtual recitations and office hours during COVID-19.

Committee Member — Caltech Student-Faculty Conference

Oct 2020 – Jan 2021

- Designed and analyzed a student survey on the chemistry curriculum.
- Collaborated with faculty on proposed revisions.

Social Director — Caltech Chemistry Club

Oct 2019 – Dec 2020

- Organized professional, outreach, and social events.
- Recruited a distinguished guest speaker for an annual symposium.

Volunteer — Caltech RISE Program

June 2019 – July 2020

- Tutored underserved high-school students in SAT prep.

SCHOLARSHIPS & AWARDS

- **Paul & Daisy Soros Fellowship for New Americans** (2025)
- **NSF Graduate Research Fellowship** (2025)
- **Goldwater Scholar in Mathematics, Science, and Engineering** (2021)
- ACS Undergraduate Award in Physical Chemistry — Caltech (2024)

- Perpall Speaking Competition Semifinalist — Caltech (2020)
- NCAA ITA Scholar-Athlete — Caltech (2019, 2020)
- Polish National Alliance Scholarship (2020, 2021, 2023)
- John Kopczynski Scholarship (2019, 2020, 2023)
- Richard Gorecki Scholarship (2020, 2023)
- National Merit Scholar (2018–2020, 2023)

OTHER EXPERIENCE

Community Member — Talon Voice

Nov 2022 – Present

- Active participant in the hands-free coding community.
- Contributed to software enabling:
  - **Voice-coding LaTeX**
  - **Dictation-LLM integration**

Student-Athlete — NCAA Tennis, Caltech

Aug 2018 – Dec 2020

- Member of Top-25 nationally ranked team.

LANGUAGES

Polish — Fluent